

2021 10%: selected excess 1 at a 67 MeV signal hypothesis

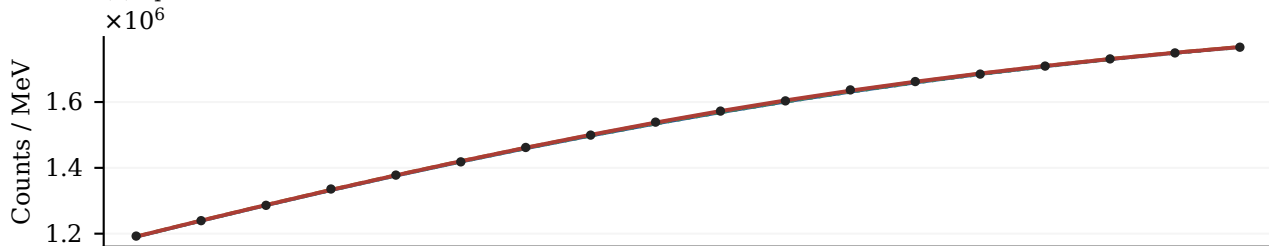
v16 neighboring signal-MC template | $c = 65.551$ MeV, $s = 1.581$ MeV

Signed full-selected yield = $44,252 \pm 16,175$; local asymptotic $Z = 2.74$ σ , $p = 0.00311$

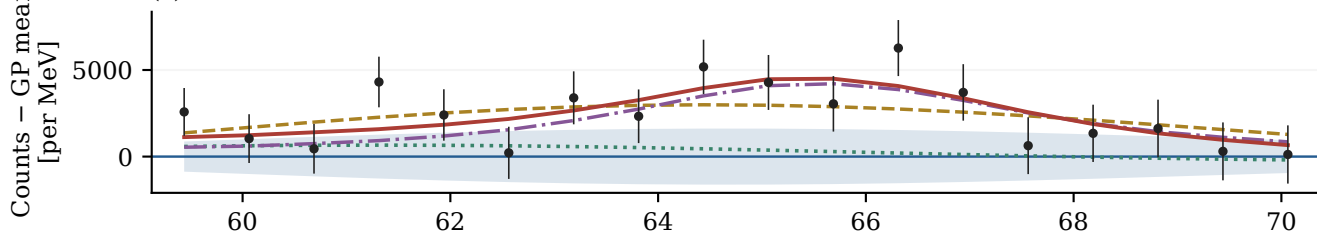
GP background-only toys: local excess rank $p = 0.002$ (1 / 1000 more extreme toys); count 95% interval [2.53e-05, 0.00556]

- Observed counts
- ⋯ Background in signed S+B fit
- ⬜ Blind region: excluded from GP training
- ⬜ GP mean and ± 1 SD constraint
- Signed signal + background
- Search mass hypothesis
- Background-only fit
- Signed signal (middle panel)
- Signal core center

(a) Spectrum in the fit interval



(b) Residuals relative to the GP mean



(c) Standardized residuals and blind window

